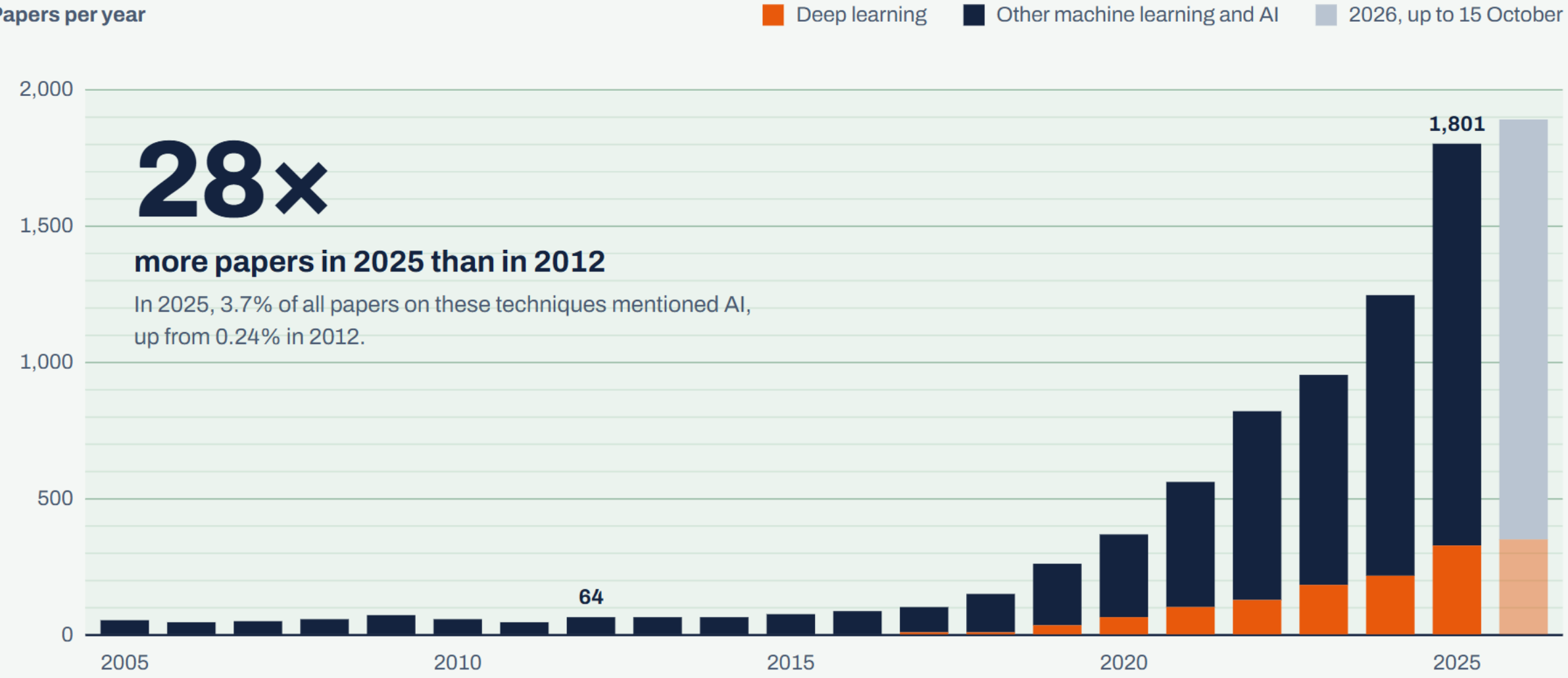
Papers per year



Mass spectrometry

788

papers in 2025, up from 39 in 2012

Chromatography

474

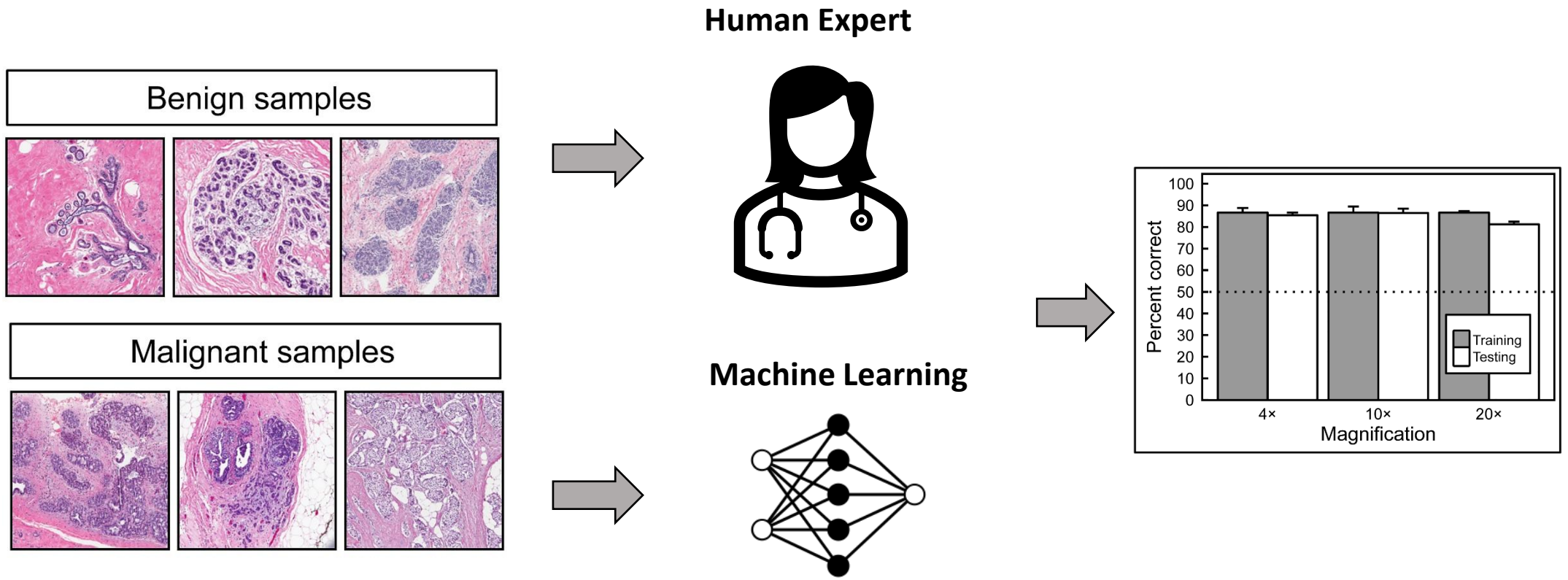
papers in 2025, up from 24 in 2012

Raman, infrared and NMR spectroscopy

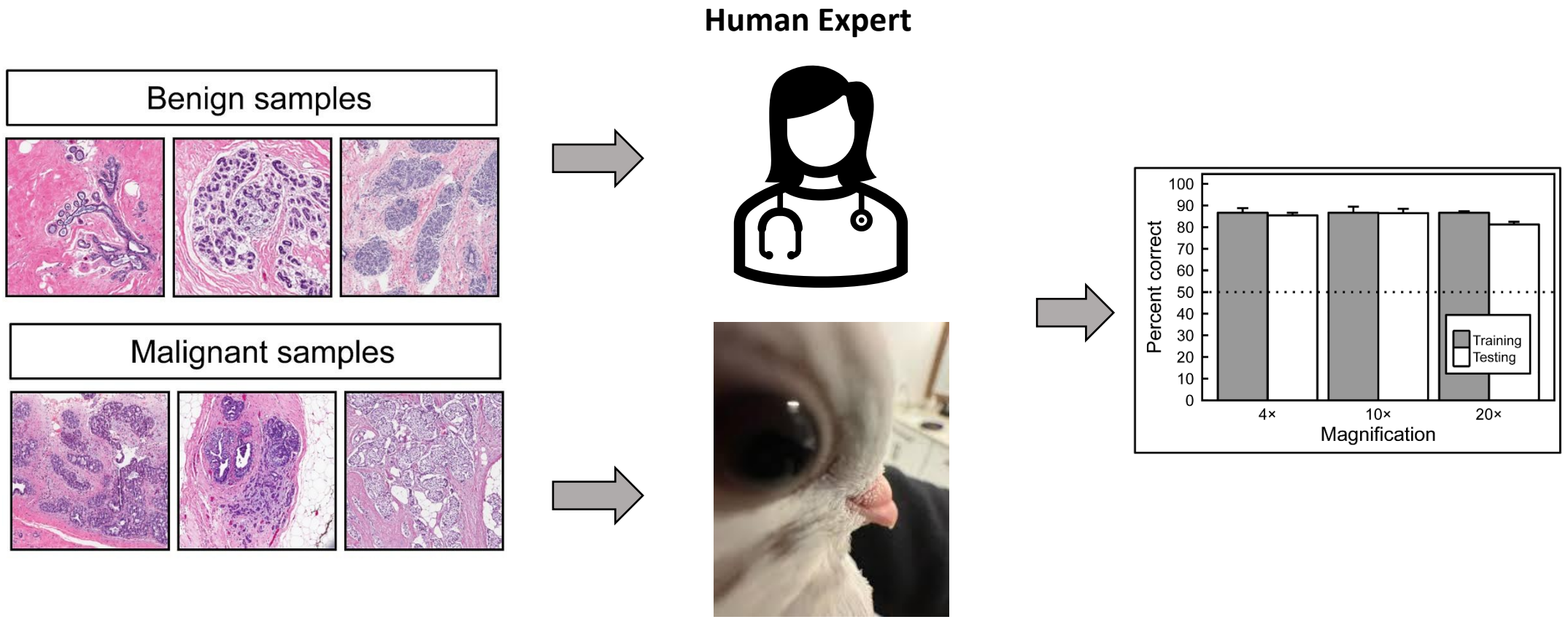
890

papers in 2025, up from 12 in 2012

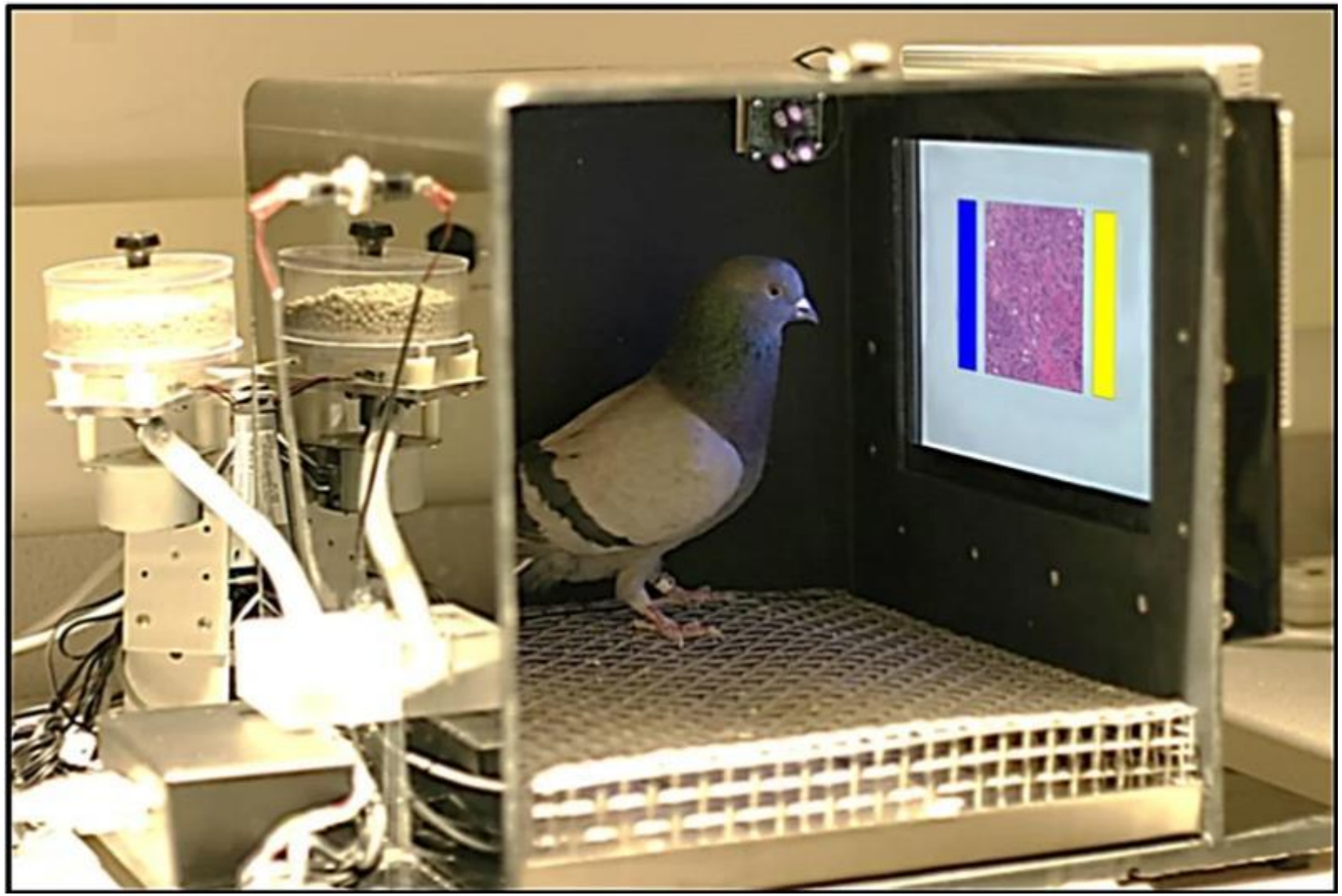
Human expert competing with Artificial Intelligence (Machine Learning model) to classify benign or malignant breast cancer slides



I lied; multiple pigeons learned to do this in just 14 days. Machine Learning works with the same principles though



ML trains a model from experience (data) with respect to a task and performance measure that should be able to generalize



Using Pigeons to Diagnose Cancer | Scientific American

ML trains a model from experience (data) with respect to a task and performance measure that should be able to generalize

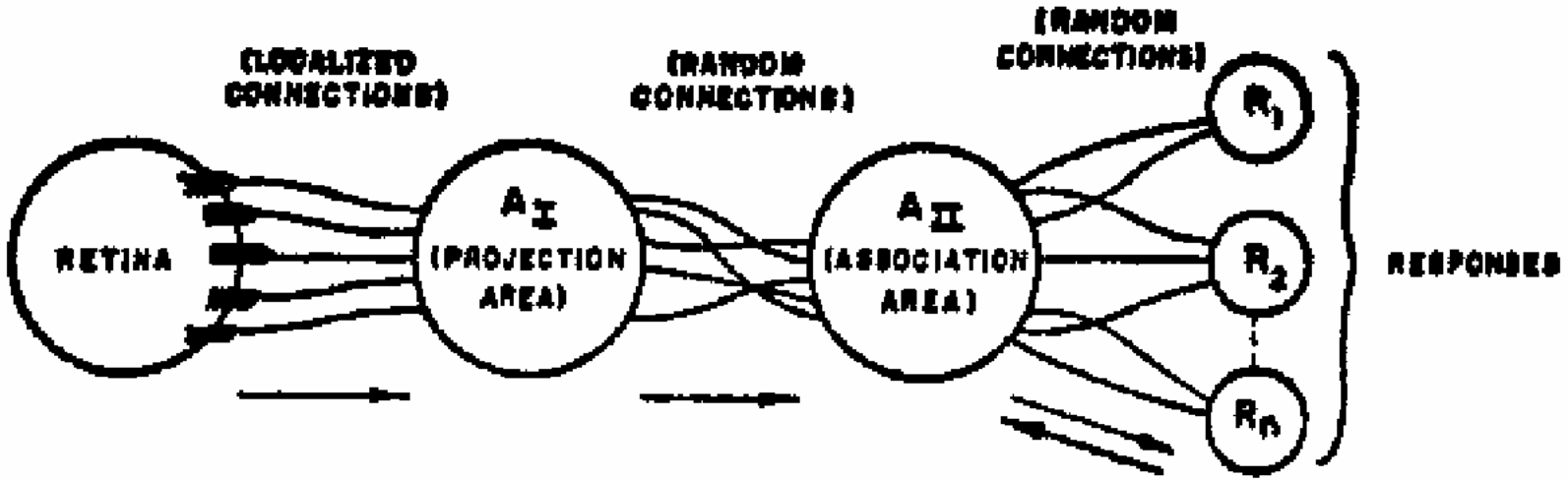


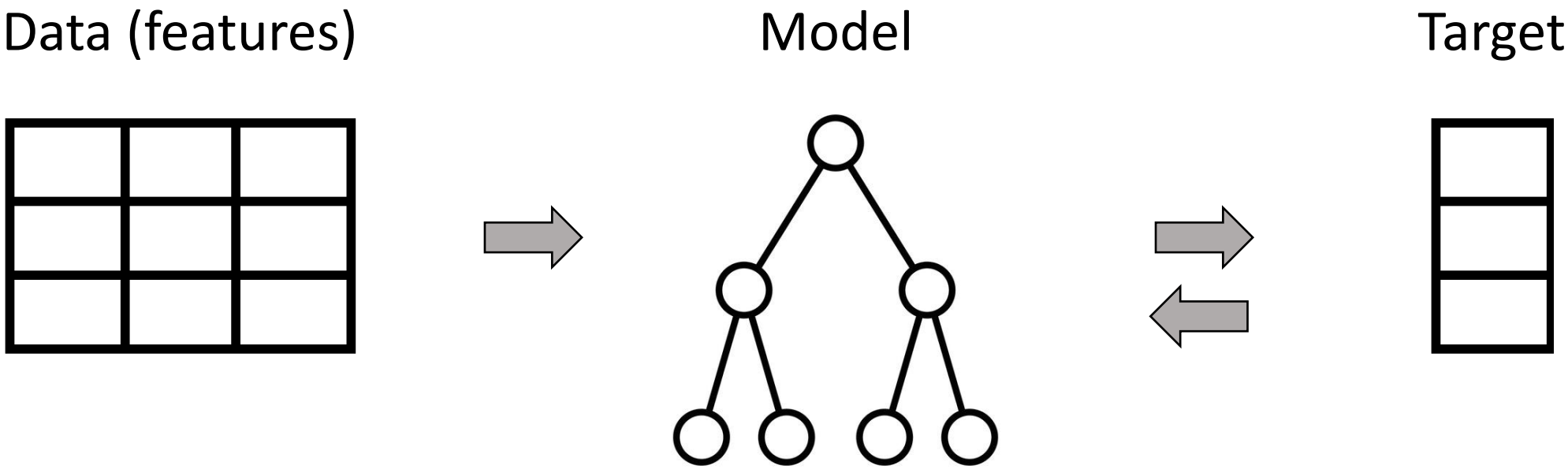
FIG. 1. Organization of a perceptron.

Rosenblatt, F. (1958). The perceptron: a probabilistic model for information storage and organization in the brain. Psychological review, 65(6), 386.

“field of study that gives computers the ability to learn without being explicitly programmed.”
Arthur Samuel - 1958

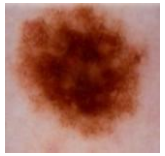
“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. ~ $P(T, E)$”
Tom M. Mitchell - 1997

Machine learning maps between data (features) and a target value, from a trained model you can obtain predictions when providing the model with only data and an unknown target

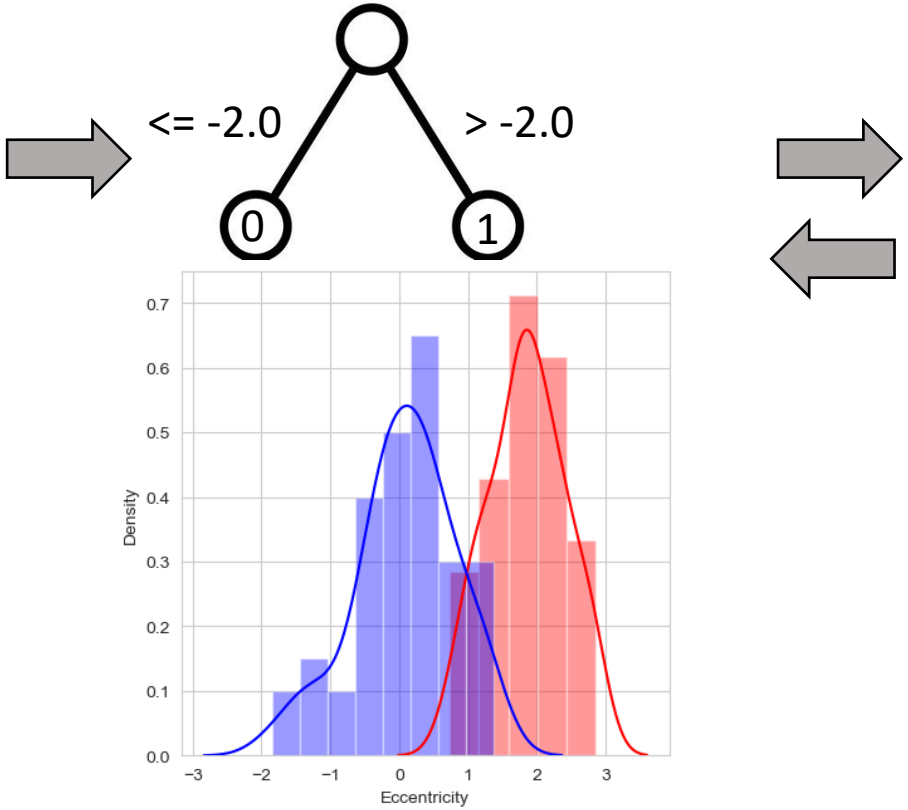


Skin melanoma as a simplified example of how a model learns; note that a computer only works with numbers!

Data (features)

	Eccentricity
	2
	-0.5
	1.8

Model

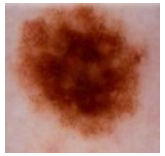


Target

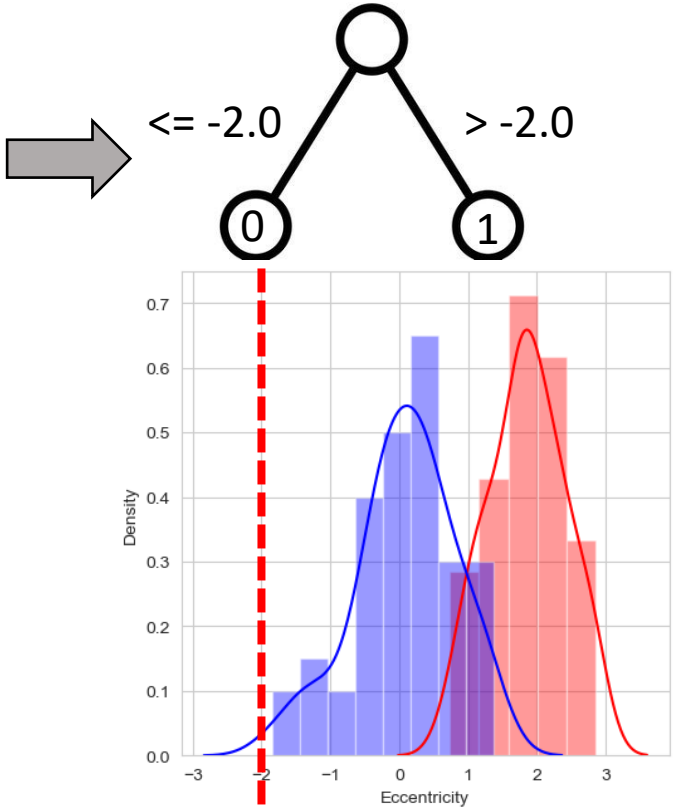
1
0
1

A random decision boundary to start, the difference between predictions and target are used to move the boundary

Data (features)

	Eccentricity
	2
	-0.5
	1.8

Model



Predictions

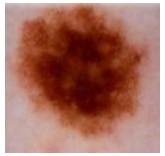
1
1
1

Target

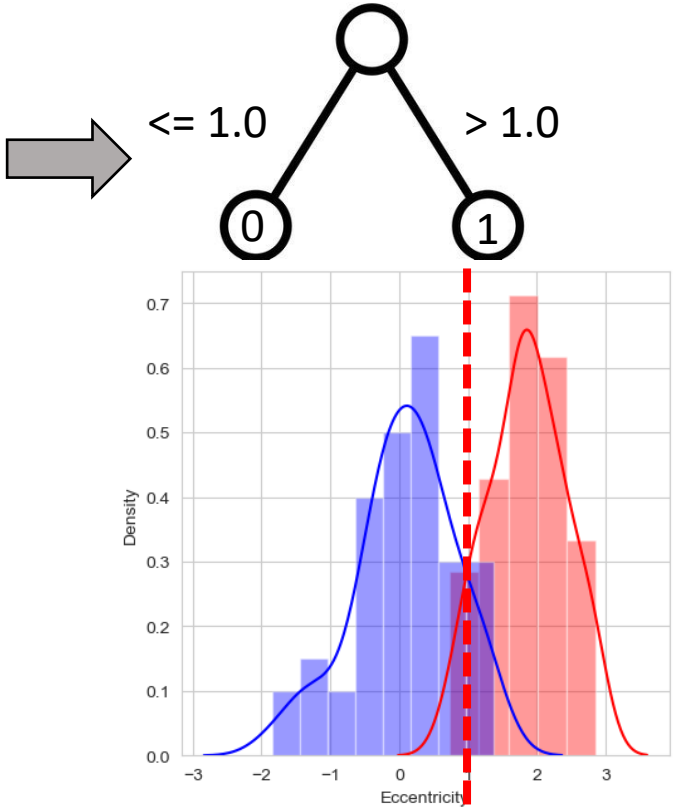
1
0
1

A decision boundary is optimized; based on presented data for a task and performance measure that should be able to generalize

Data (features)

	Eccentricity
	2
	-0.5
	1.8

Model



Predictions

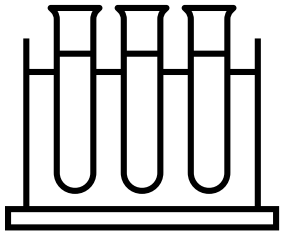
1
0
1

Target

1
0
1

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



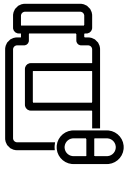
Medical imaging



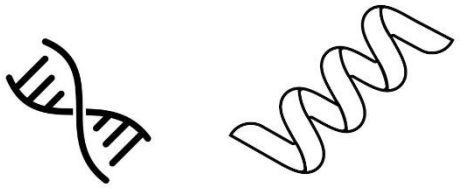
ECG

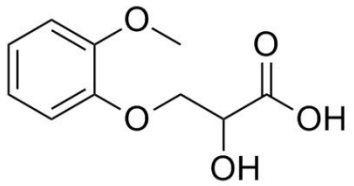


Drug design



Omics



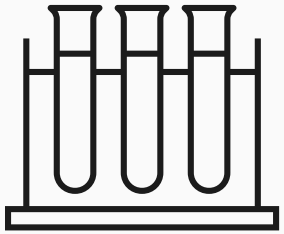


Medical records



AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



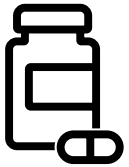
Medical imaging



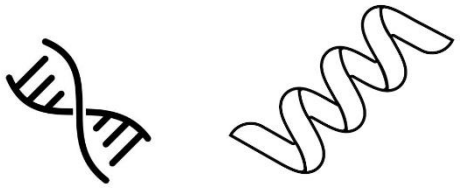
ECG

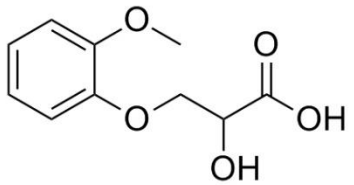


Drug design



Omics





Medical records



Type of data
- Tabular

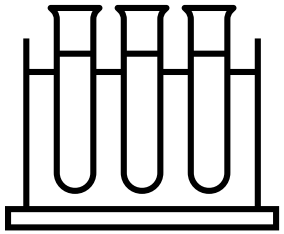
Type of models
- Tree-based
- Continuous
- Neural-network (deep learning)

Target
- Disease state
- Anomaly detection

Ning et al. Scientific Reports (2025); Lauritsen et al. Artificial Intelligence in Medicine (2020)

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



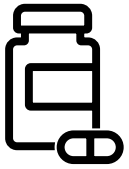
Medical imaging



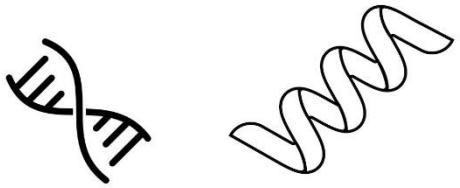
ECG

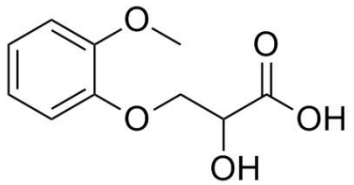


Drug design



Omics





Medical records



Type of data

- Images/videos

Type of models

- Tree-based
- Continuous
- Convolutional Neural-network (deep learning)

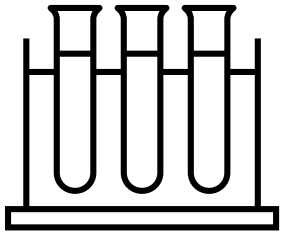
Target

- Disease state
- Segmentation
- Anomaly detection

Le Dinh et al. Applied Sciences (2022)

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



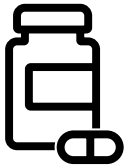
Medical imaging



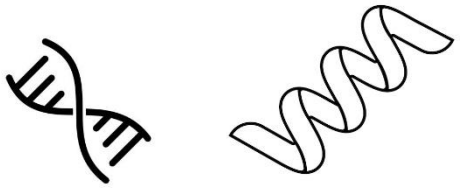
ECG

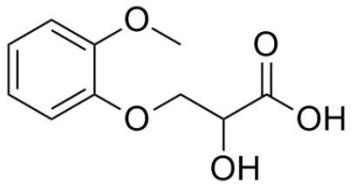


Drug design



Omics





Medical records



Type of data

- Temporal ordered (sequential)

Type of models

- Tree-based
- Continuous
- Recurrent Neural-network (deep learning)

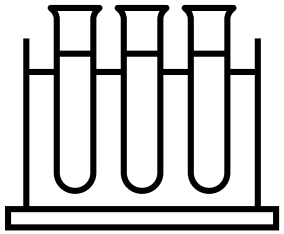
Target

- Disease state
- Anomaly detection

Le Dinh et al. Applied Sciences (2022)

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



Medical imaging



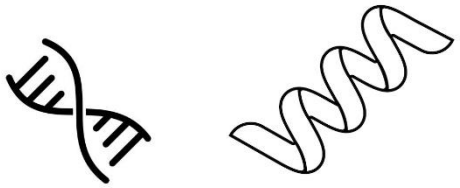
ECG

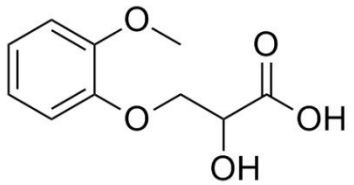


Drug design



Omics





Medical records



Type of data
- Tabular
- Structural (2D/3D)

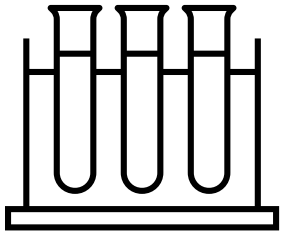
Type of models
- Tree-based
- Continuous
- Recurrent/Convolutional/Graph Neural-networks (deep learning)

Target
- Drug efficiency
- IC50
- Drug stability

Arus-Pous et al. Journal of Cheminformatics (2020)

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



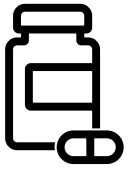
Medical imaging

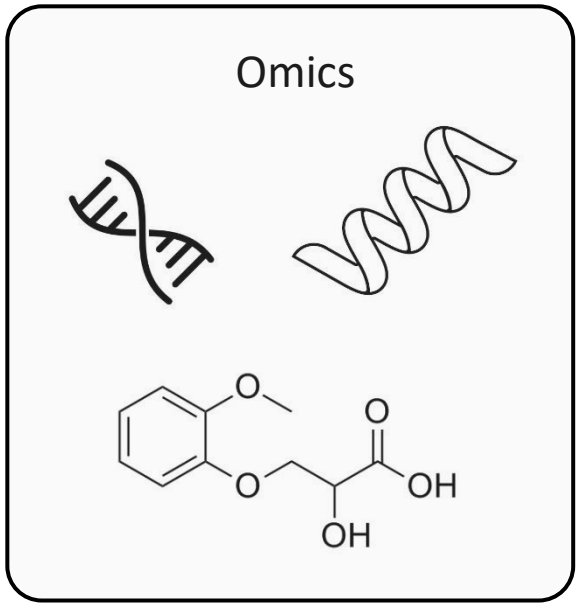


ECG



Drug design





Medical records



Type of data
- Tabular
- Sequential

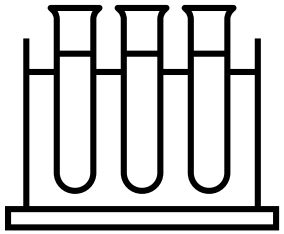
Type of models
- Tree-based
- Continuous
- Recurrent Neural-networks (deep learning)

Target
- Pathogenicity
- Structure
- Interactions

Jumper et al. Nature (2021)

AI has an impact on many different healthcare topics; from small molecules to medical records

Biomedical analytics



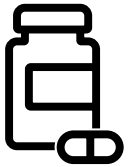
Medical imaging



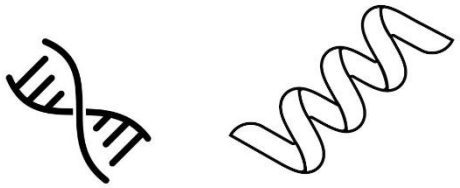
ECG



Drug design

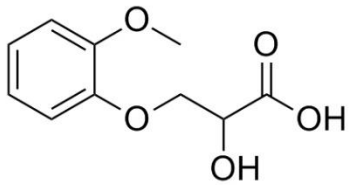


Omics



Medical records





Type of data

- Human written text

Type of models

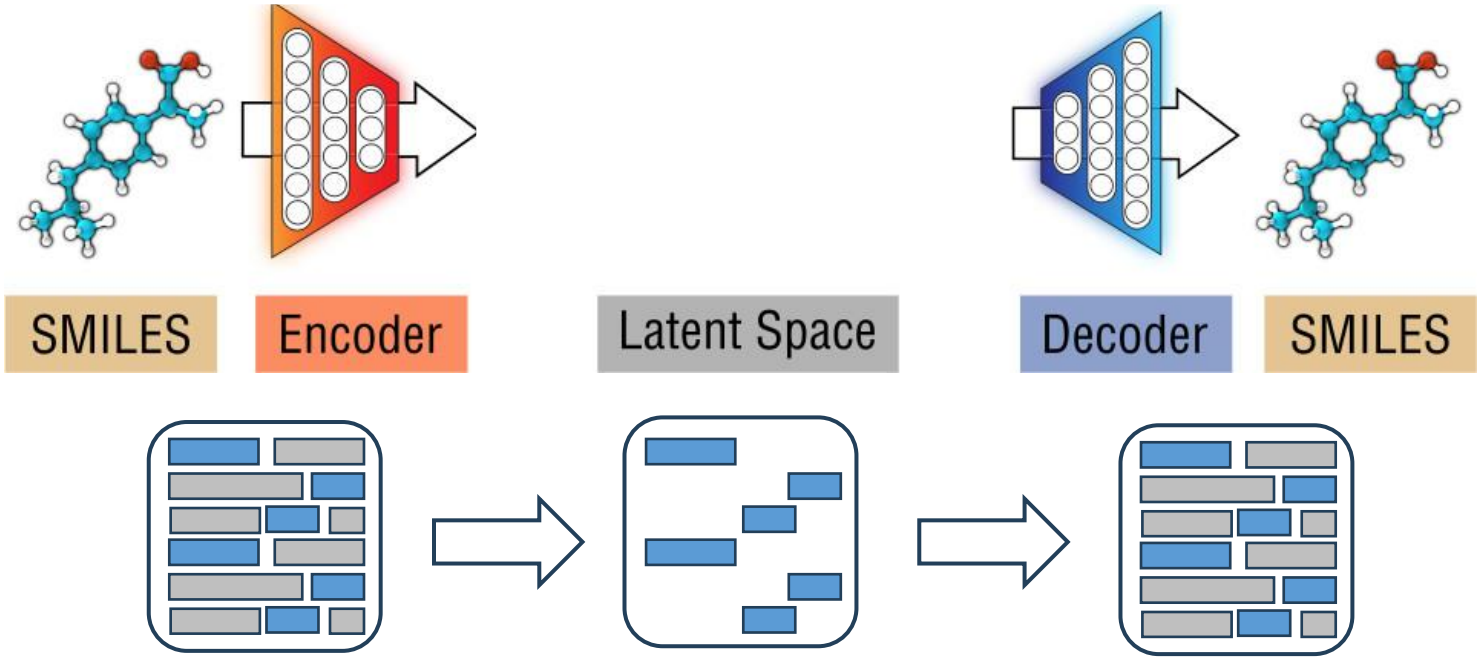
- Transformer Neural-networks (deep learning)
- Recurrent Neural-networks (deep learning)

Target

- Model the underlying textual structure

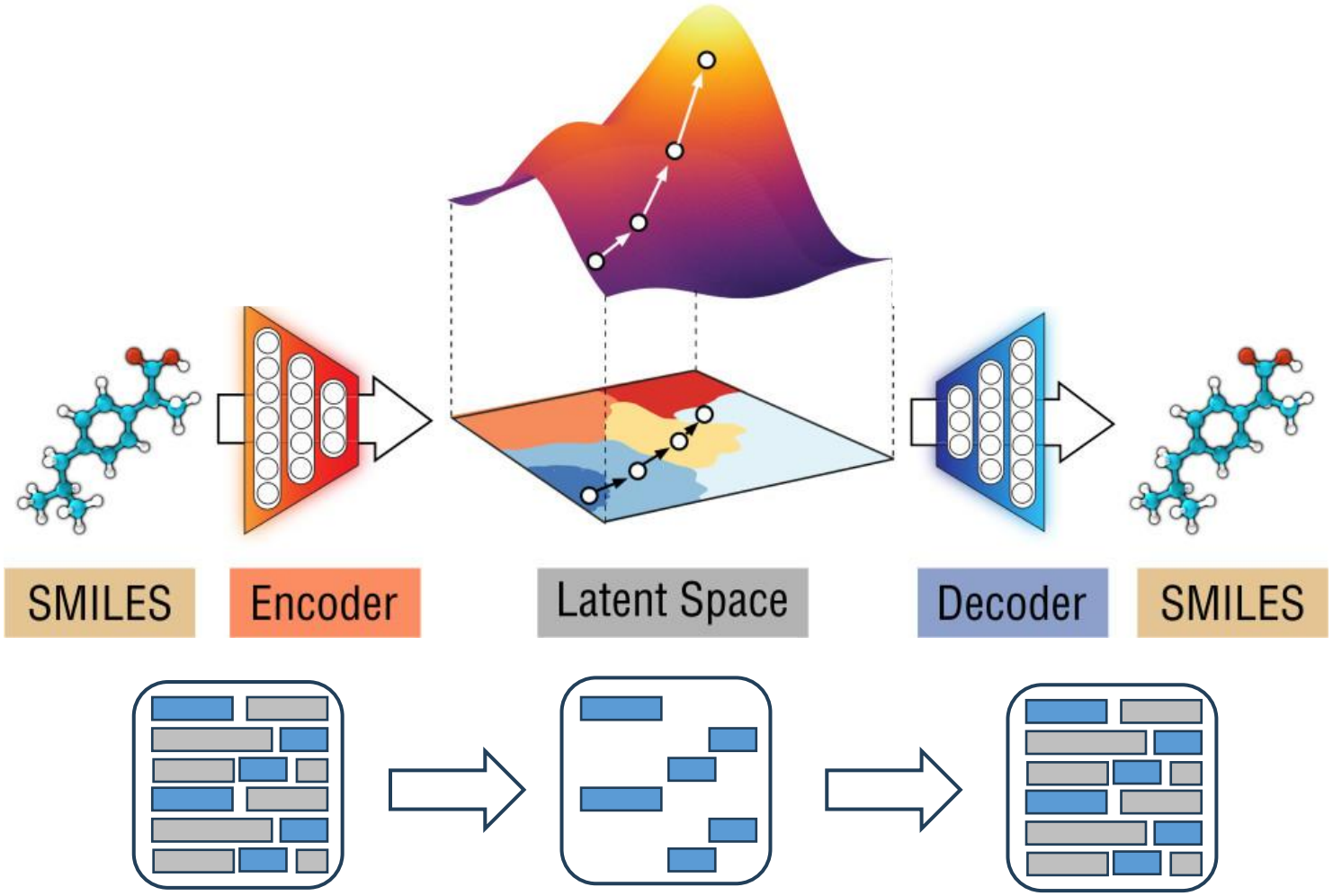
Yao Jiang et al. Nature (2023)

Drug design with variational auto-encoders; able to sample the latent space to find potentially interesting drugs



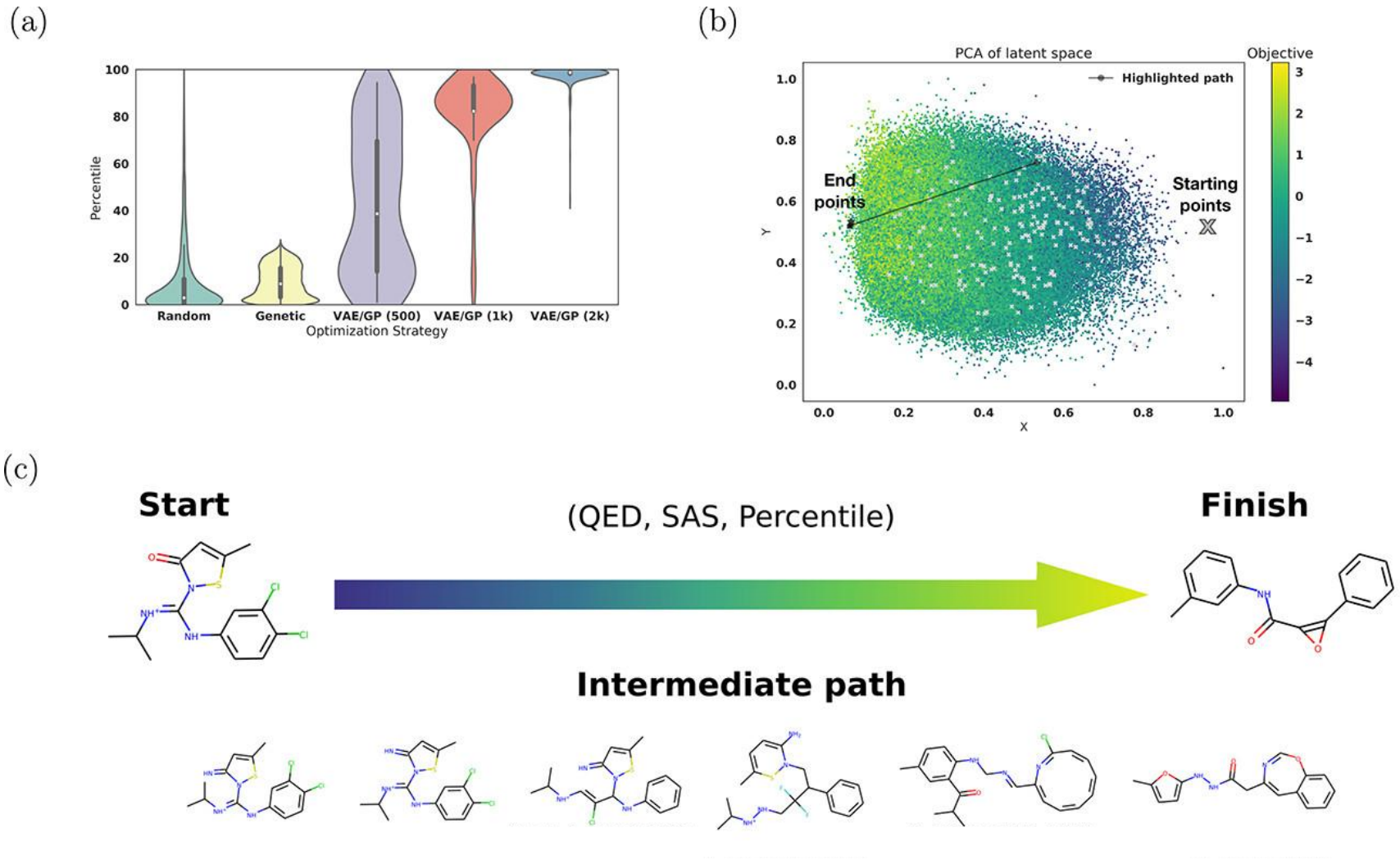
Gomez-Bombarelli et al. ACS Central Science (2023)

Drug design with variational auto-encoders; able to sample the latent space to find potentially interesting drugs



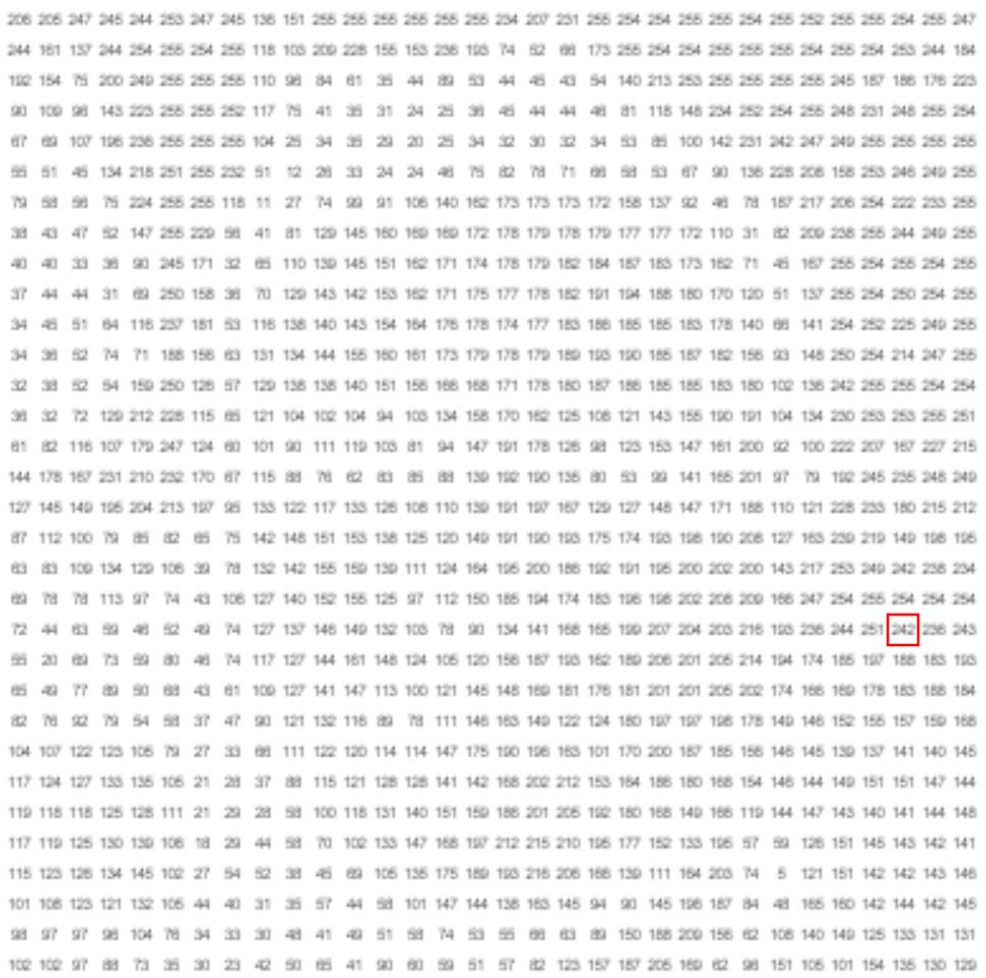
Gomez-Bombarelli et al. ACS Central Science (2023)

Reconstructing ibuprofen-like molecules yields much higher percentile (likeness) scores for the variational autoencoder



Gomez-Bombarelli et al. ACS Central Science (2023)

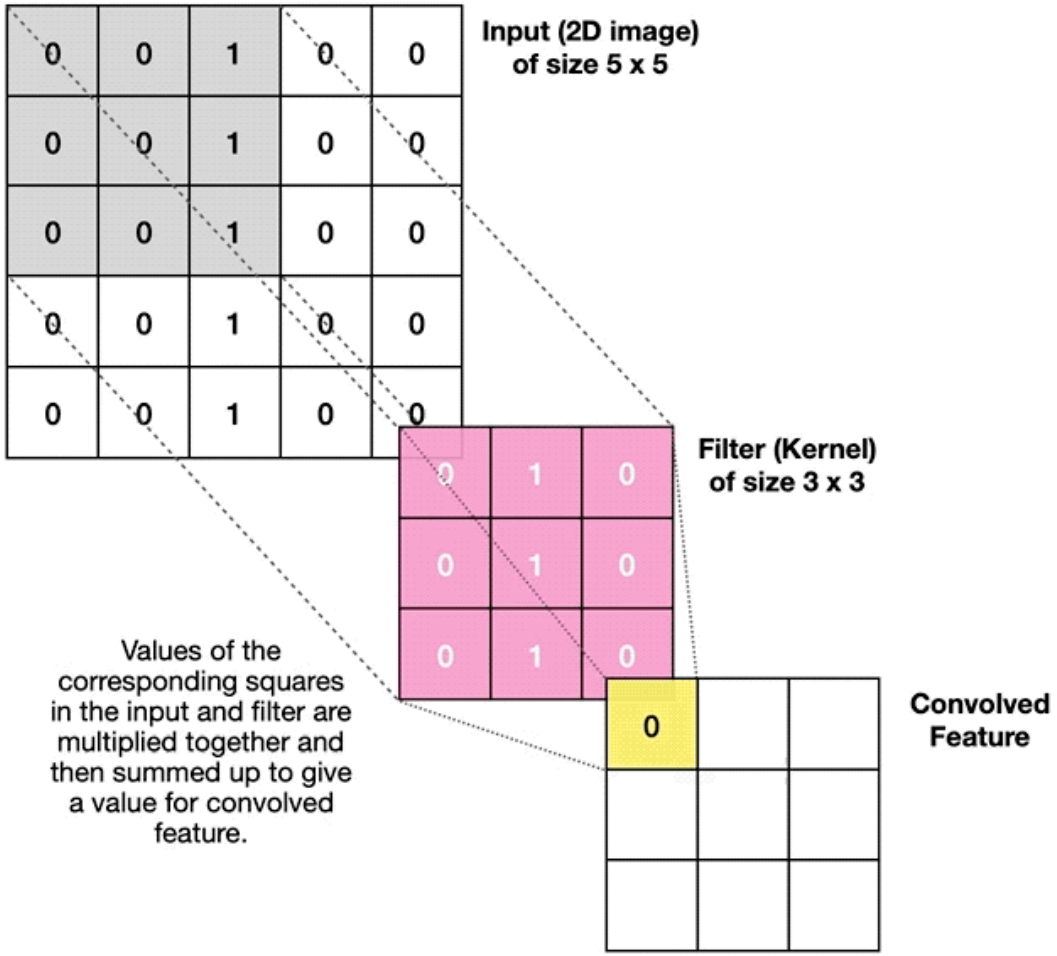
Computer vision is a large subfield in AI that deals with 2D and 3D images and videos



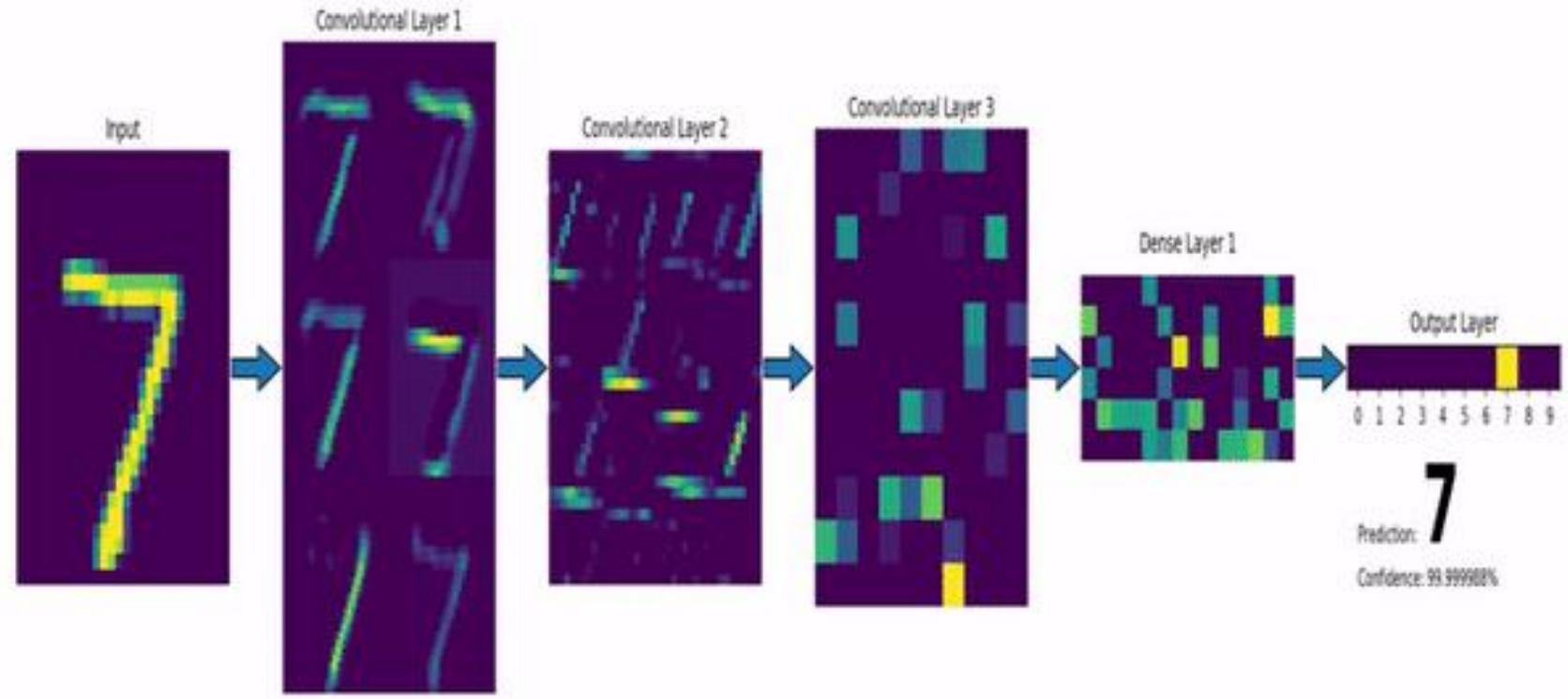


https://setosa.io/ev/image-kernels/

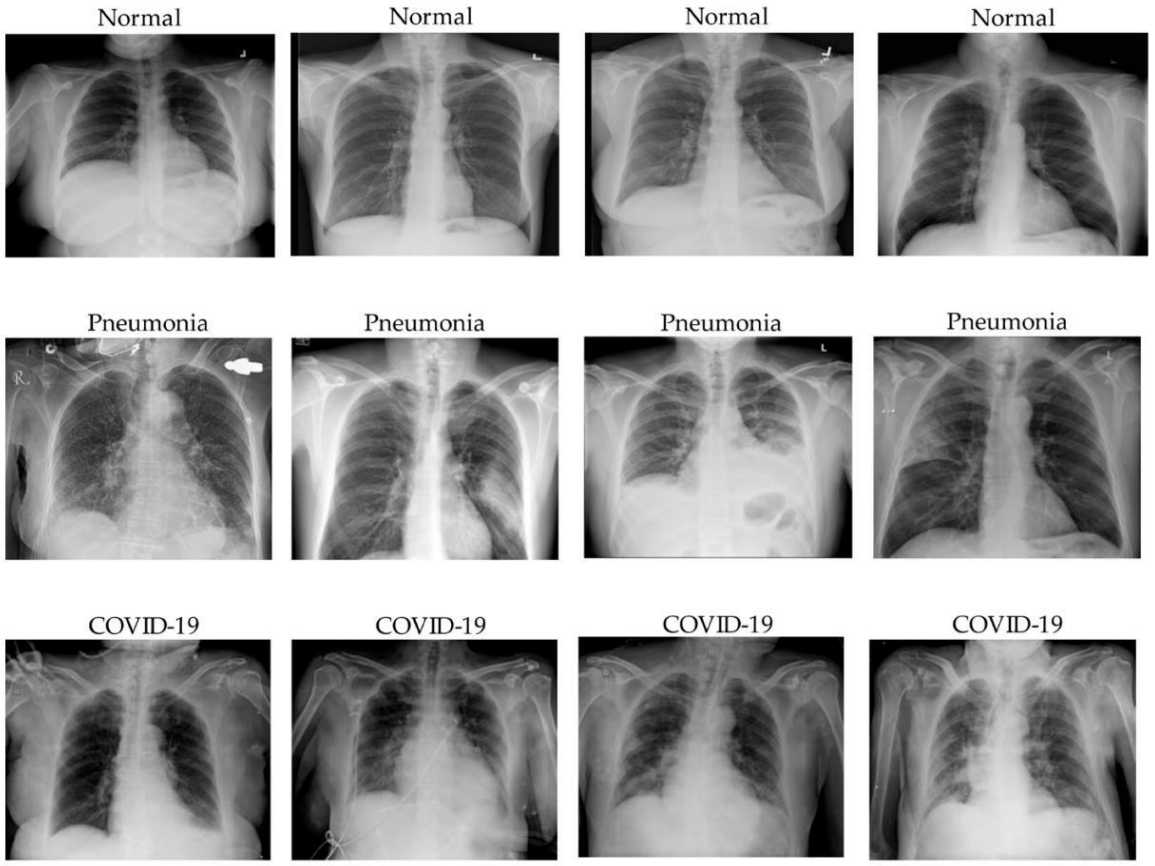
Filters are used to obtain small and unlocalized features of images that can be combined in subsequent convolutional layers

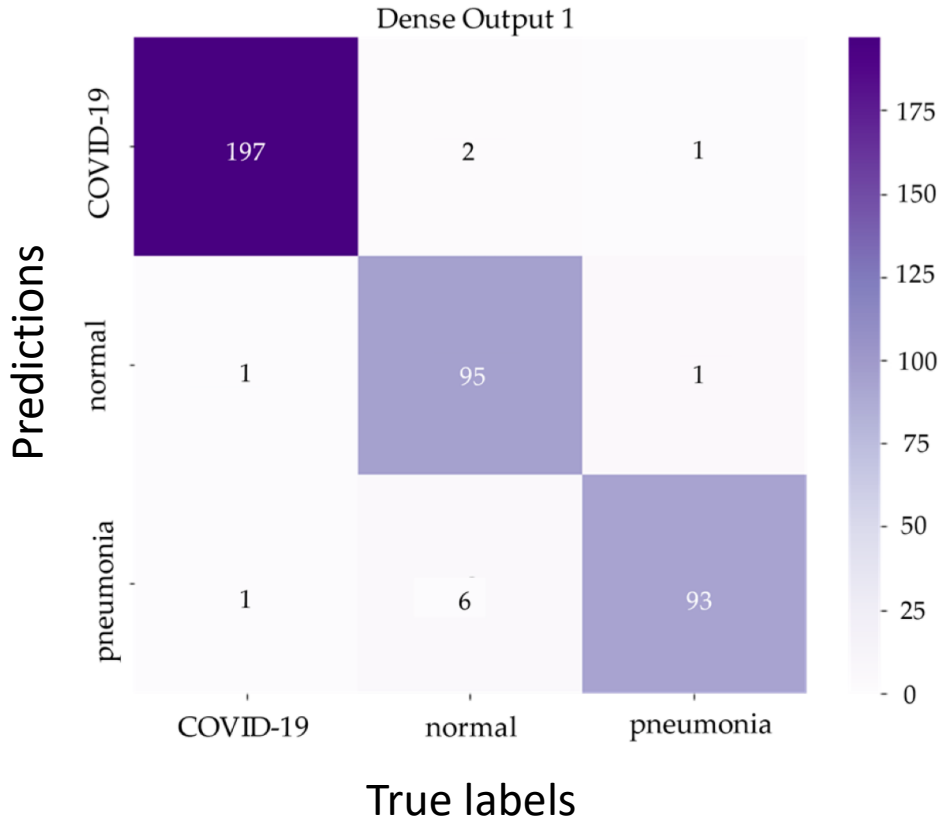


Filters are used to obtain small and unlocalized features of images that can be combined in subsequent convolutional layers



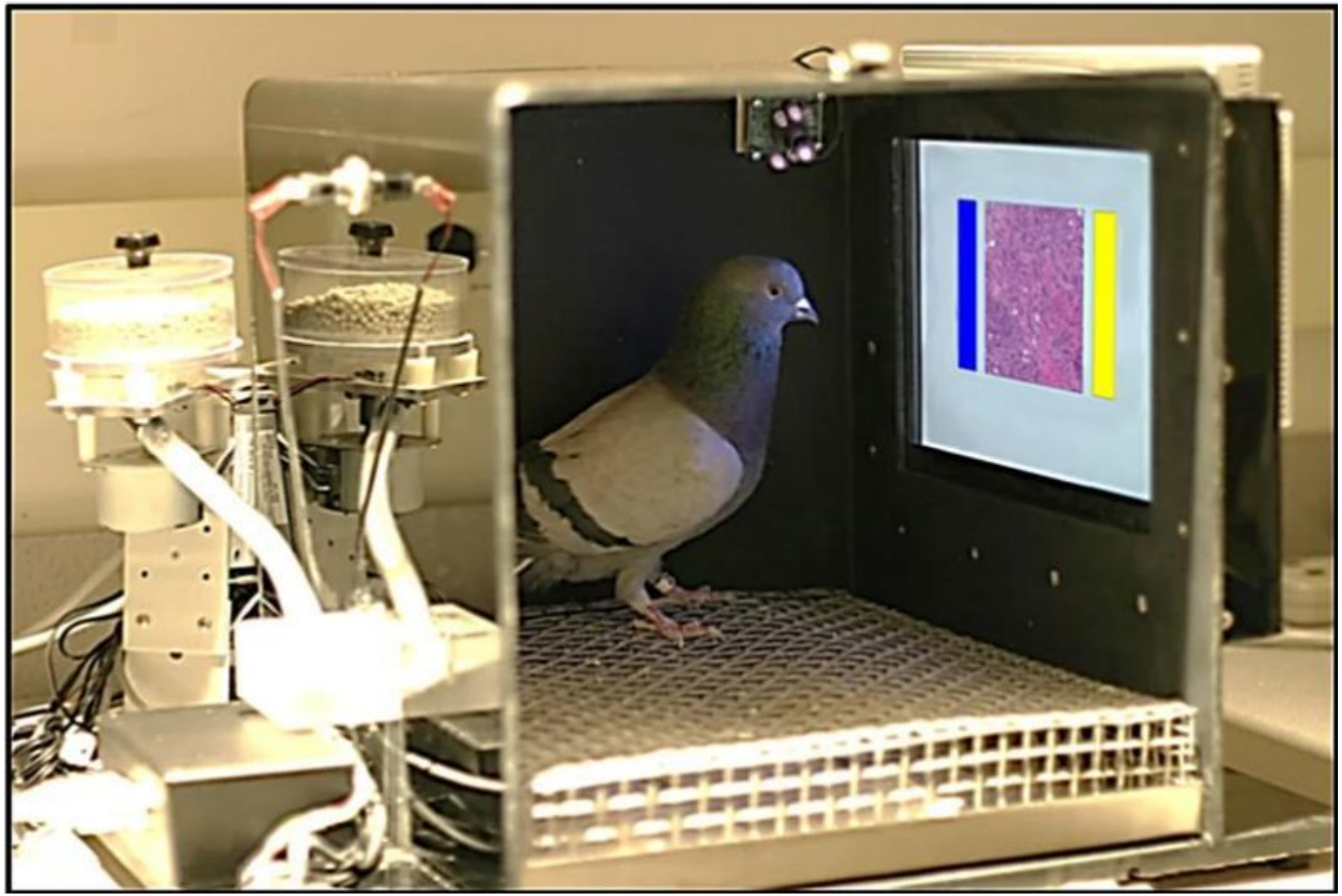
COVID can be detected with high sensitivity and specificity, and also be distinguished from pneumonia





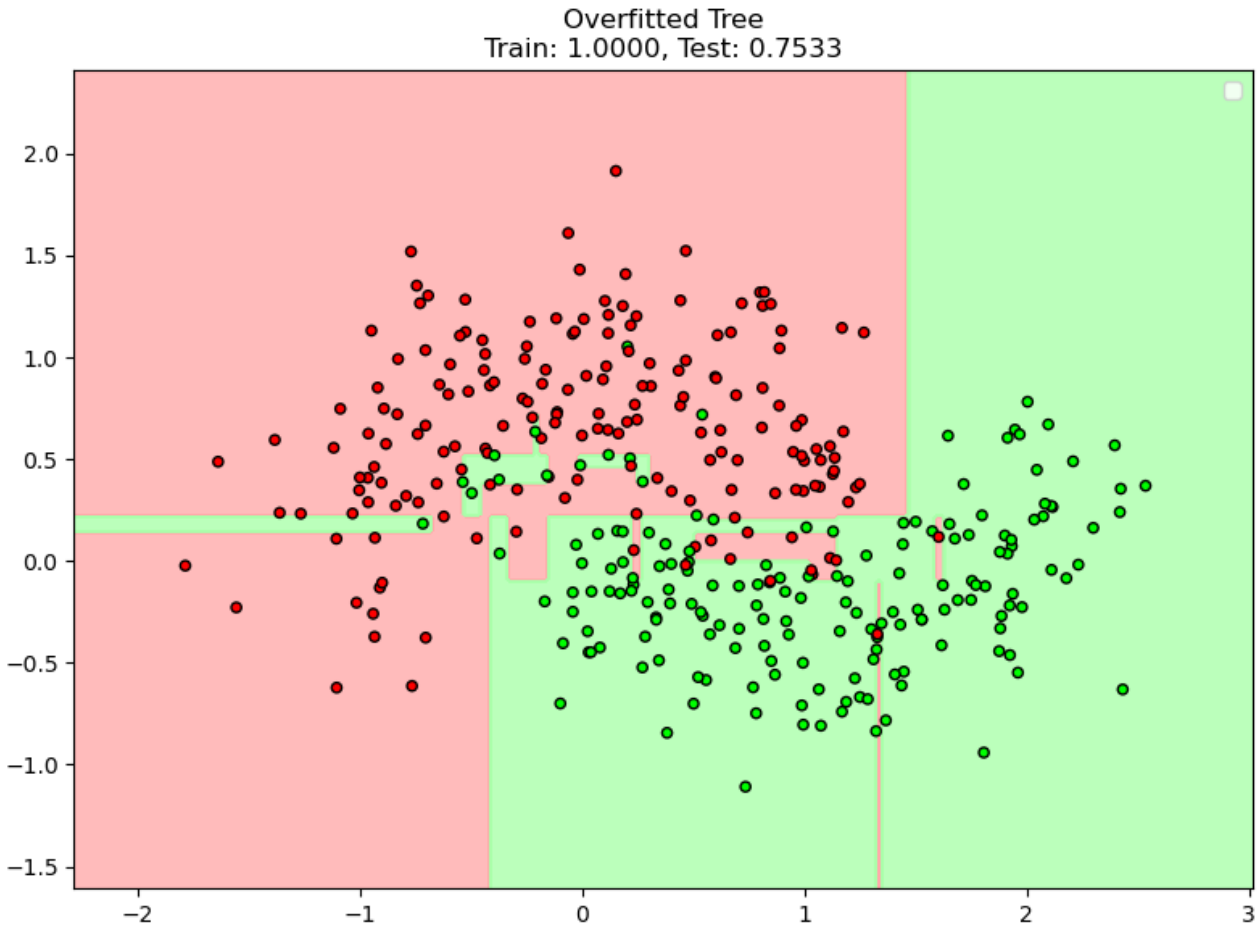
Le Dinh, T, et al. COVID-19 Chest X-ray Classification and Severity Assessment Using Convolutional and Transformer Neural Networks. Appl. Sci. 2022, 12, 4861

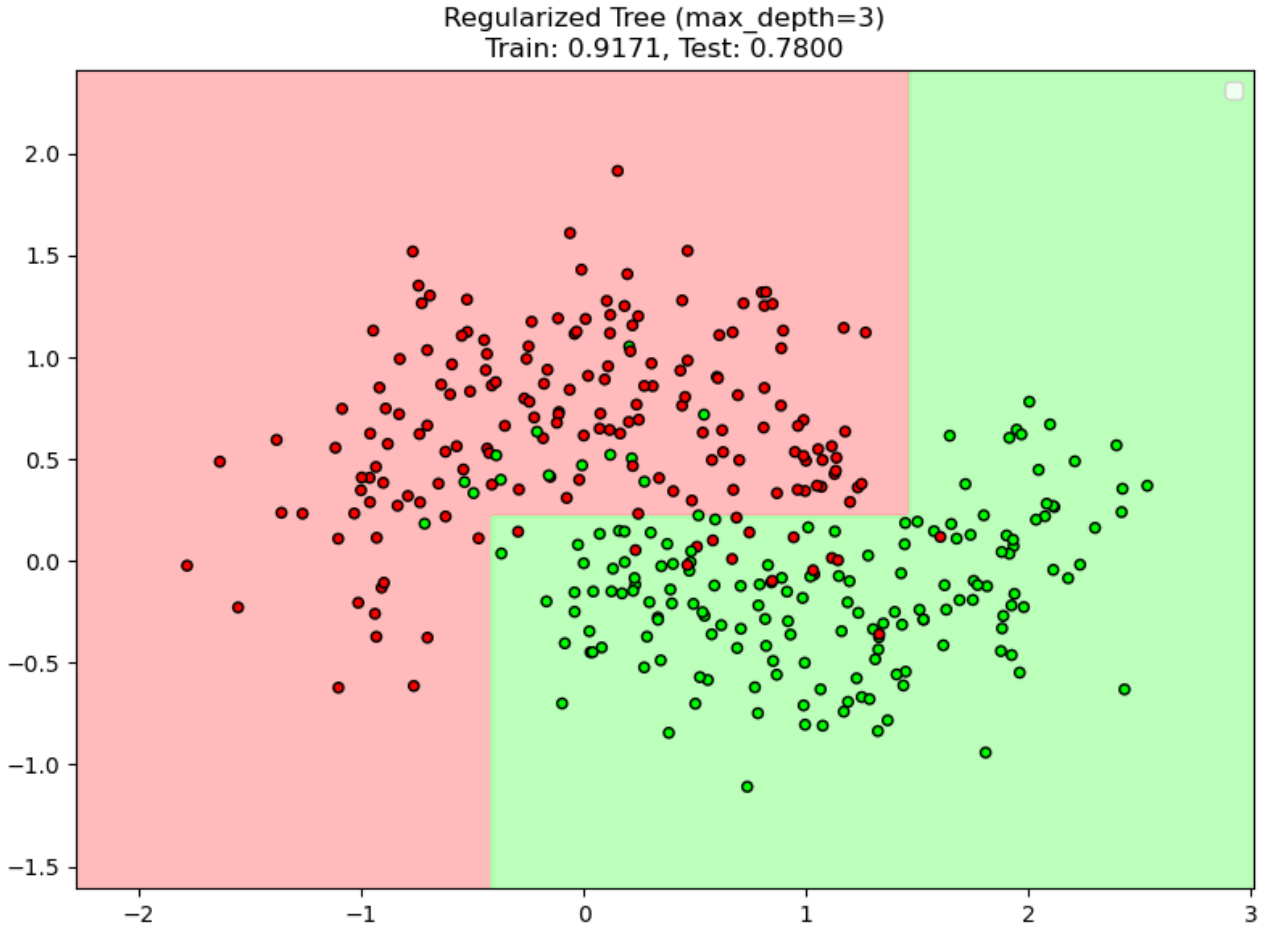
ML trains a model from experience (data) with respect to a task and performance measure that should be able to generalize



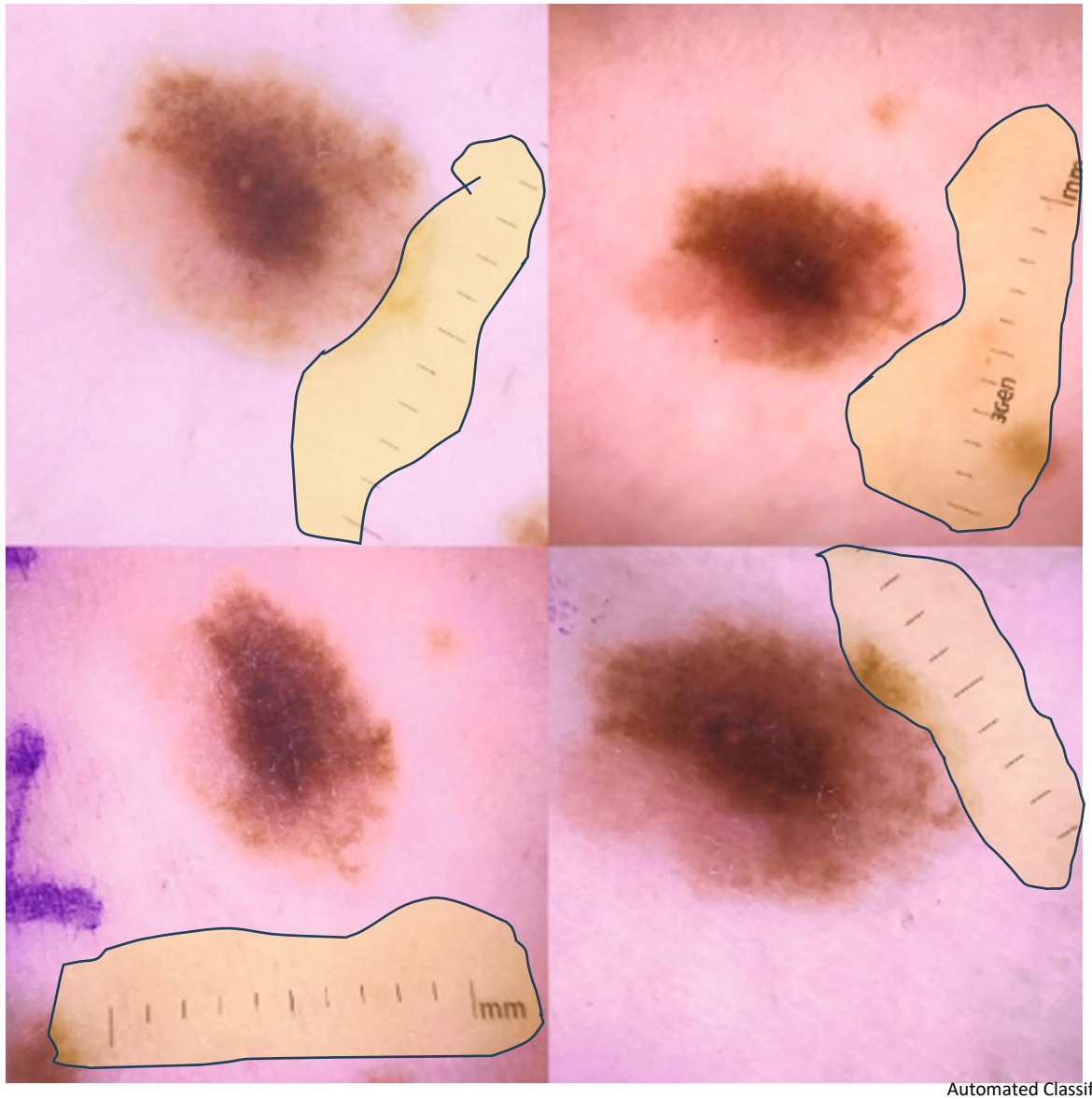
Using Pigeons to Diagnose Cancer | Scientific American

AI cheating


Overfitted Tree
Train: 1.0000, Test: 0.7533
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2
3

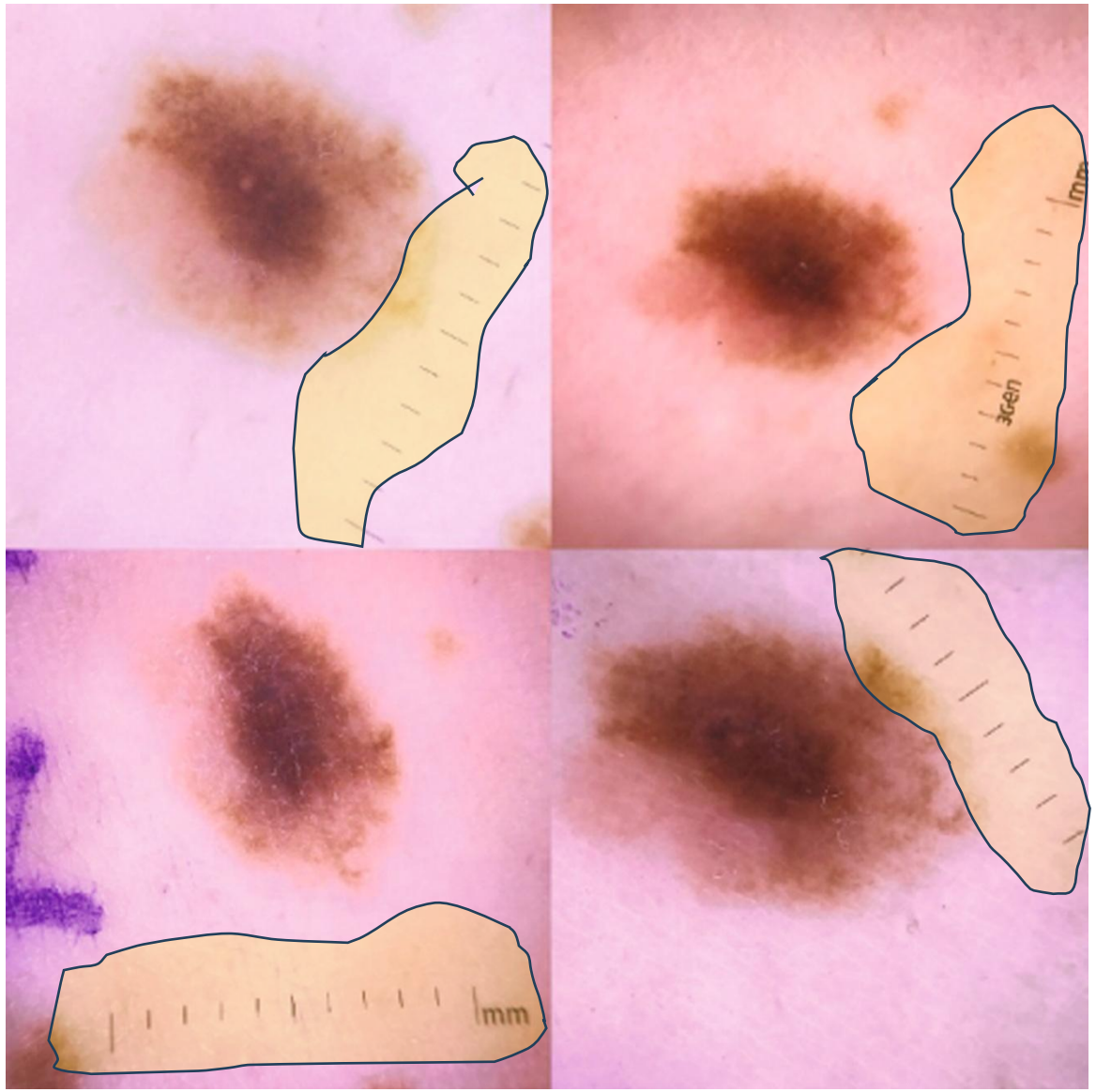

Regularized Tree (max_depth=3)
Train: 0.9171, Test: 0.7800
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2
−1
0
1
2
3

AI cares about optimizing the objective function; AI can cheat

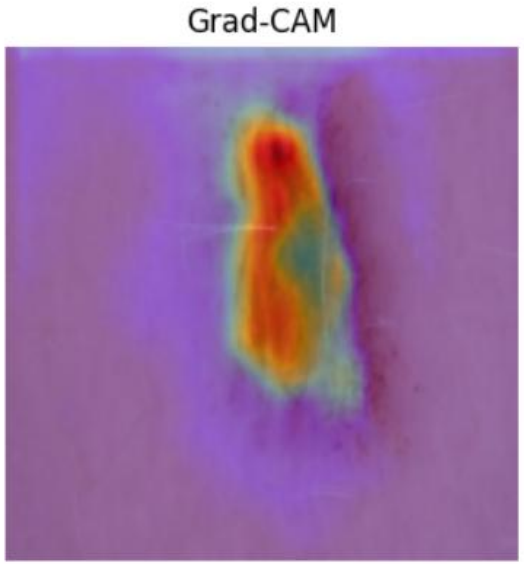


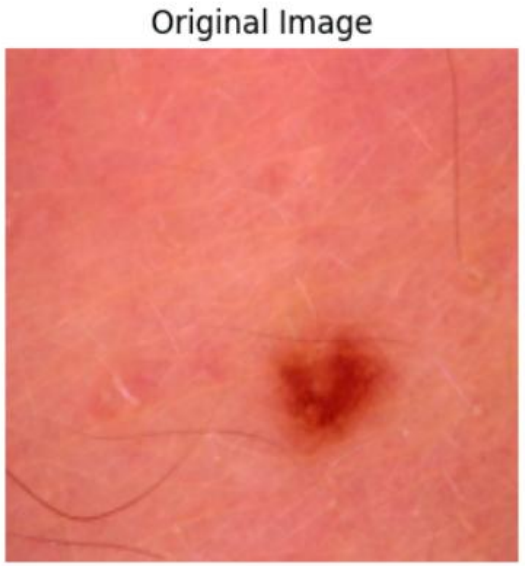
Automated Classification of Skin Lesions: From Pixels to Practice - Akhila Narla · Brett Kuprel · Kavita Sarin · Roberto Novoa · Justin Ko

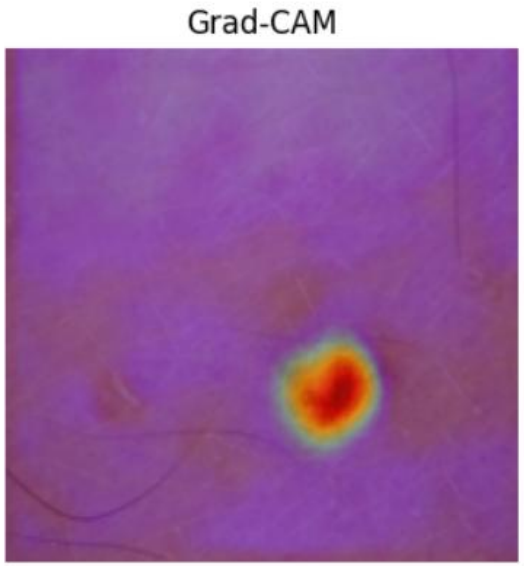
AI cares about optimizing the objective function; AI can cheat











Conclusion and outlook

ML trains a model from experience (data) with respect to a task and performance measure that should be able to generalize

Different objectives, ML algorithms, and input data:

- disease state, anomaly detection, segmentation, drug efficiency, IC50, drug stability, pathogenicity, structure, interactions, textual structure, …
- tree-based, continuous, neural-network, recurrent NN, convolutional NN, transformer (LLM), …
- tabular, image, video, temporal ordered, sequence, structural, textual, …